ECA08 – ANALOG AND DIGITAL COMMUNICATION

CO1 Engineering Design Report

AM-Based Communication Link Design: From Modulation Selection to Receiver Deployment

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Flow of the report: REQUIREMENT → ALTERNATIVES → ANALYSIS → SELECTION → DESIGN → REALIZATION → VERIFICATION → DEPLOYMENT DECISION

Summary of the design (details and proofs follow)

Modulation: conventional AM (DSB-FC), μ = 0.7, envelope detection. **Occupied bandwidth:** 8 kHz inside a 20 kHz slot (12 kHz guard).

Power: P_c = 10 W, sidebands 2.45 W, P_T = 12.45 W, efficiency 19.7 %, peak envelope power 28.9 W. **Channels:** 53 ideal (52 on a grid containing 1 MHz).

Receiver: superheterodyne, IF = 455 kHz, f_LO = 1455 kHz, image = 1910 kHz, image rejection ≈ 36.8 dB (single tuned stage, Q = 50).

Decision: YES for community voice broadcasting; NO for analog-TV use of this chain (TV needs VSB and a 6 MHz channel).

Stage 1 – Interpret the Communication Design Requirement

1.1 Basic communication-system block diagram

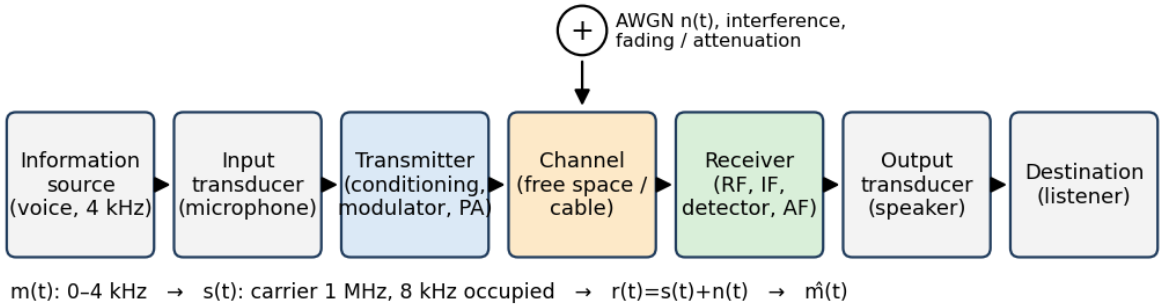


Fig. 1 Basic communication system with the design signals marked

The **source** produces the 0–4 kHz voice; the **input transducer** converts sound to an electrical signal m(t). The **transmitter** conditions m(t), generates a carrier, modulates it and amplifies it. The **channel** attenuates the signal and adds AWGN n(t) plus interference from neighbouring stations. The **receiver** selects the wanted station, rejects the others and demodulates; the **output transducer** reproduces sound. The design target is to recover m(t) with acceptable SNR and fidelity.

1.2 Need for modulation and limits of baseband transmission

- **Antenna size:** an efficient antenna is about λ/4. For a 4 kHz signal λ = c/f = 75 km, so λ/4 ≈ 18.75 km. At 1 MHz, λ = 300 m and λ/4 = 75 m (loaded or ferrite antennas make this practical).
- **No multiplexing:** every voice signal occupies the same 0–4 kHz, so stations would overlap. Modulation shifts each station to its own carrier (frequency-division multiplexing).
- **Noise/interference and range:** baseband signals suffer more from low-frequency interference and cannot be selected by a tuned circuit; a high-frequency carrier propagates and is easily filtered.
- **Hardware:** tuned circuits and amplifiers are easier to build at a fixed carrier with a narrow fractional bandwidth (8 kHz / 1 MHz = 0.8 %).

1.3 Bandwidths (f_m = 4 kHz)

Scheme	Message BW	Theoretical transmission BW	Reason
Baseband	4 kHz	4 kHz	No translation
Conventional AM	4 kHz	2 f_m = 8 kHz	Carrier + upper + lower sideband

Scheme	Message BW	Theoretical transmission BW	Reason
DSB-SC	4 kHz	2 f_m = 8 kHz	Both sidebands, no carrier
SSB-SC	4 kHz	f_m = 4 kHz	One sideband only
VSB-SC	4 kHz	f_m + f_v ≈ 4 + 1 = 5 kHz	One sideband + vestige f_v (1 kHz assumed)

1.4 RF channel planning and number of channels

Stations share the band by **frequency-division multiplexing**: each is given a carrier and the carriers are spaced at least by the occupied bandwidth plus a guard band, so spectra do not overlap and a receiver can pick one by tuning.

$$N = (1600 - 540)\ kHz\ /\ 20\ kHz = 1060\ /\ 20 = 53\ channels\ (ideal)$$

If 1 MHz must itself be a carrier, the grid is 560, 580, …, 1580 kHz (each slot ±10 kHz stays inside 540–1600 kHz) which gives **52 usable channels**; the 53rd needs the grid shifted by 10 kHz.

1.5 AM power (P_c = 10 W, μ = 0.7)

$$P_T = P_c\,(1 + \mu^2/2) = 10\,(1 + 0.245) = 12.45\ W$$

$$P_{SB} = P_c\,\mu^2/2 = 2.45\ W\ \ (1.225\ W\ per\ sideband);\quad \eta = P_{SB} / P_T = 2.45 / 12.45 = 19.7\ \%$$

So 80.3 % of the transmitted power is spent in the carrier, which carries no information. The peak envelope power is $P_c(1+\mu)^2 = 10 \times 2.89 =$ **28.9 W**; the power amplifier must be rated for this peak to stay linear.

1.6 Receiver requirements

Parameter	Meaning	Requirement for this link
Sensitivity	Smallest input giving acceptable output SNR	≈ 2 μV (50 Ω) for 20 dB output SNR, NF = 10 dB (Sec. 4.5)
Selectivity	Reject adjacent stations	Pass ±4 kHz; ≥ 40 dB at ±16 kHz (nearest adjacent sideband edge)
Fidelity	Faithful audio 0–4 kHz	Passband droop < 1 dB at 4 kHz; low distortion at μ = 0.7
Stability	Oscillator / IF drift	Drift much less than ±5 kHz IF window; crystal or PLL LO
Image rejection	Reject f_s + 2 f_IF	≥ 30–40 dB from RF preselector
IF	Fixed processing frequency	Chosen for image rejection vs achievable IF filter Q (Sec. 4.4)

1.7 Practical constraints of the educational prototype

- **Power:** a 10 W transmitter on a licensed band is not legal or necessary in a laboratory. The 10 W figure is used for design calculations; the prototype runs at milliwatt level into a dummy load or short wire coupling (follow local radio regulations).
- **Components:** low-cost ICs (MC1496/AD633 modulator), 455 kHz ceramic filters, ferrite-rod antenna and a ganged variable capacitor; component tolerance of 5–10 % limits tracking accuracy.
- **Test equipment:** function generator, oscilloscope and spectrum analyser or sound-card FFT; AWGN is emulated with a noise generator or in software.

Stage 1 deliverable – Communication Design Requirement Statement

Bandwidth: message 4 kHz; AM occupied BW 8 kHz inside a 20 kHz slot; 53 ideal channels in 540–1600 kHz.

Power: P_c = 10 W, μ = 0.7 → P_T = 12.45 W, sidebands 2.45 W, PEP 28.9 W. **Multiplexing:** FDM, 20 kHz carrier spacing, ≥ 12 kHz guard band.

Receiver: sensitivity ≈ 2 μV, ≥ 40 dB adjacent-channel rejection, fidelity to 4 kHz, image rejection ≥ 30 dB, stable IF; low-power educational implementation.

Stage 2 – Investigate and Select the Modulation Scheme

2.1 Time- and frequency-domain expressions

Scheme	Time domain	Frequency domain / remarks
AM (DSB-FC)	s(t) = A_c [1 + μ m(t)] cos ω_c t, |μ m| ≤ 1. Tone: A_c cosω_c t + (A_cμ/2) cos(ω_c+ω_m)t + (A_cμ/2) cos(ω_c−ω_m)t	S(f) = (A_c/2)[δ(f−f_c)+δ(f+f_c)] + (A_cμ/4)[M(f−f_c) +M(f+f_c)]. Carrier line + 2 sidebands; BW = 2W

Scheme	Time domain	Frequency domain / remarks
DSB-SC	$s(t) = A_c\, m(t) \cos \omega_c t$	$S(f) = (A_c/2)[M(f-f_c)+M(f+f_c)]$. No carrier line; BW = 2W
SSB-SC	$s(t) = (A_c/2)[m(t) \cos \omega_c t \mp \hat{m}(t) \sin \omega_c t]$ (− : USB, + : LSB), $\hat{m}$ = Hilbert transform	One sideband only; BW = W
VSB-SC	$s(t) = [A_c\, m(t) \cos \omega_c t] * h_v(t) = (A_c/2)[m \cos \omega_c t \mp m_q \sin \omega_c t]$	$S(f) = (A_c/2)[M(f-f_c)+M(f+f_c)]\, H(f)$; needs $H(f-f_c)+H(f+f_c) = \text{const}$ for $\lvert f \rvert \le W$. BW = W + f_v

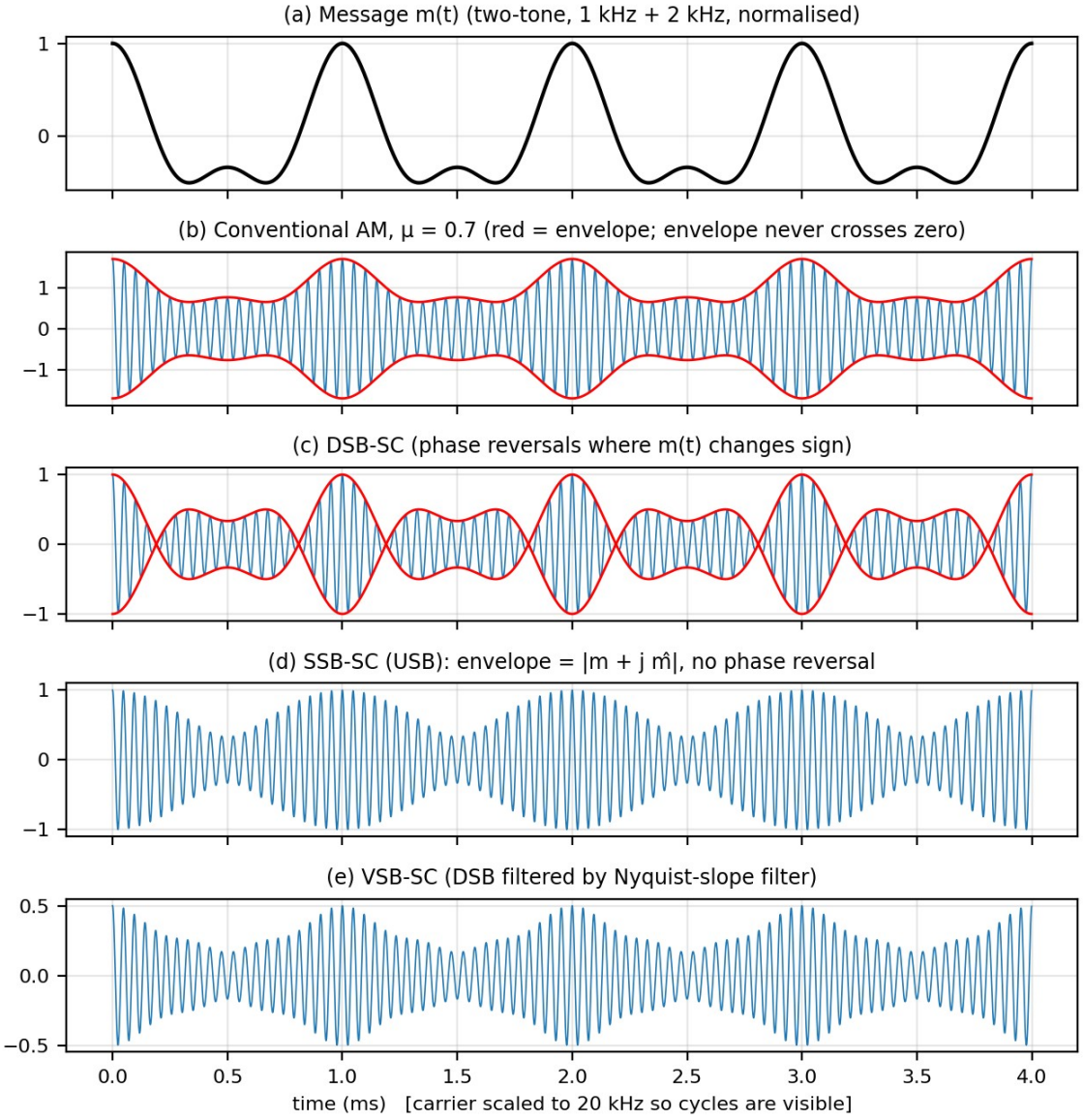


Fig. 2 Generated waveforms of the four schemes (Python; carrier scaled to 20 kHz for visibility)

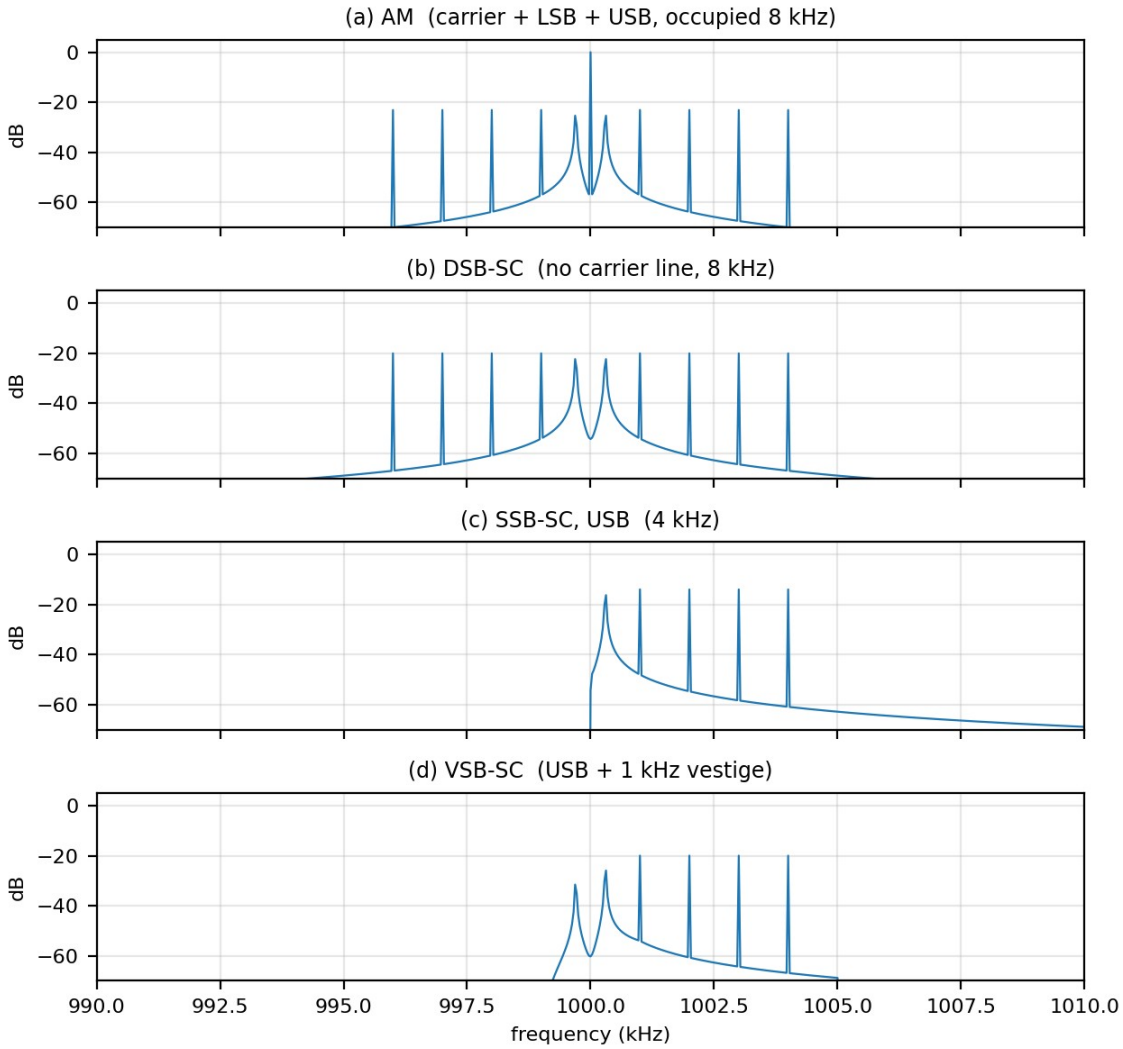


Fig. 3 Spectra at f_c = 1 MHz (message tones 0.3, 1, 2, 3, 4 kHz; Python FFT)

Interpretation: the AM spectrum shows the carrier line plus two mirror-image sidebands (8 kHz wide); DSB-SC shows the same sidebands without the carrier; SSB keeps one 4 kHz sideband; in VSB the 300 Hz tone appears in both sidebands but each is partly attenuated by the Nyquist slope (gains 0.35 and 0.65), so their sum is still the full amplitude.

2.2 Power comparison and quantified advantage

Quantity	AM	DSB-SC	SSB-SC
Carrier power	10 W	0 W (suppressed)	0 W
Sideband power transmitted	2.45 W	2.45 W	1.225 W (one sideband)
Total transmitted power	12.45 W	2.45 W	1.225 W
Power saved vs AM	–	10.0 W (80.3 %)	11.225 W (90.2 %)
Power ratio vs AM	1	12.45/2.45 = 5.08 (7.06 dB)	12.45/1.225 = 10.16 (10.07 dB)
Efficiency (info power / total)	19.7 %	100 %	100 %
Output SNR at equal P_T (figure of merit)	$\mu^2/(2+\mu^2)$ = 0.197 (−7.06 dB)	1 (0 dB)	1 (0 dB)

2.3 Why VSB is used for analog television, and the vestige

Video has a bandwidth of about 4.2 MHz and large energy at very low frequencies (down to d.c., the average picture brightness). DSB would need 8.4 MHz. SSB would need a filter with an almost vertical skirt at the carrier (the sideband begins at f_c itself) and would distort the low video frequencies and phase response. **VSB sends the full upper**

sideband and only a vestige (≈ 0.75 MHz) of the lower sideband: the channel shrinks to 6 MHz, the filter skirt can be gentle (easy to build, linear phase), and the low-frequency video content is kept. The vestige is complemented at the receiver by the Nyquist-slope response, which passes the carrier at 50 % amplitude so that vestige + missing part add to a constant (Fig. 7). Envelope detection of the video carrier then becomes practical with only small quadrature distortion.

2.4 Comparison of generation, detection, synchronisation and complexity

Scheme	Generation	Detection	Synchronisation	Complexity / cost
AM	Simple multiplier or collector modulation + DC offset	Diode + RC envelope detector	None	Lowest; one diode in the receiver
DSB-SC	Balanced modulator (ring, MC1496)	Product detector, coherent	Carrier phase and frequency (Costas / PLL / pilot); phase error φ gives cosφ, null at 90°	Higher; carrier recovery in every receiver
SSB-SC	Balanced mod + sharp sideband filter or phasing method	Product detector	Frequency error Δf shifts the pitch by Δf ("Donald-Duck")	High; narrow filter (≈ 300 Hz transition at 455 kHz) and sync
VSB-SC	Balanced mod + vestigial filter	Sync or envelope with Nyquist filter	Moderate	High; precise filter shape

2.5 Design Decision 1

Criterion (weight)	AM	DSB-SC	SSB-SC	VSB-SC
Receiver simplicity (30)	5	2	1	2
Receiver cost (15)	5	3	2	2
Bandwidth efficiency (20)	2	2	5	4
Power efficiency (20)	1	4	5	4
Tolerance to tuning error (15)	5	2	2	3
Weighted total (/100)	**72**	**51**	**58**	**59**

Scores 1–5 are engineering judgement; weights reflect a one-transmitter, many-receivers broadcast where receiver cost dominates. They are the assumption behind the decision, not a measured quantity.

Design Decision 1

I recommend **conventional AM (DSB-FC) with envelope detection** because: (i) its 8 kHz bandwidth fits the 20 kHz slot with a 12 kHz guard band; (ii) the receiver needs no carrier recovery, only a diode and an RC network, so listeners' sets stay cheap; (iii) it tolerates oscillator and tuning errors of kilohertz, where SSB and DSB-SC would distort; (iv) $\mu = 0.7 < 1$ avoids over-modulation with a PA rating of 28.9 W peak; (v) it scored 72/100 against 51–59 for the alternatives.

The price is **80.3 % of the power wasted in the carrier** and an output SNR 7.06 dB below DSB/SSB at equal transmitted power. SSB would be preferred only if the link were power- or spectrum-limited and the receivers could be made coherent.

Stage 3 – RF Channel Allocation and Transmitter Design

3.1 Channel plan (five representative channels)

Channel	Carrier f_c	LSB (f_c−4 … f_c)	USB (f_c … f_c+4)	Occupied band	Gap to next occupied band
1	960 kHz	956–960	960–964	956–964 kHz	12 kHz
2	980 kHz	976–980	980–984	976–984 kHz	12 kHz
3 (design)	1000 kHz	996–1000	1000–1004	996–1004 kHz	12 kHz
4	1020 kHz	1016–1020	1020–1024	1016–1024 kHz	12 kHz
5	1040 kHz	1036–1040	1040–1044	1036–1044 kHz	–

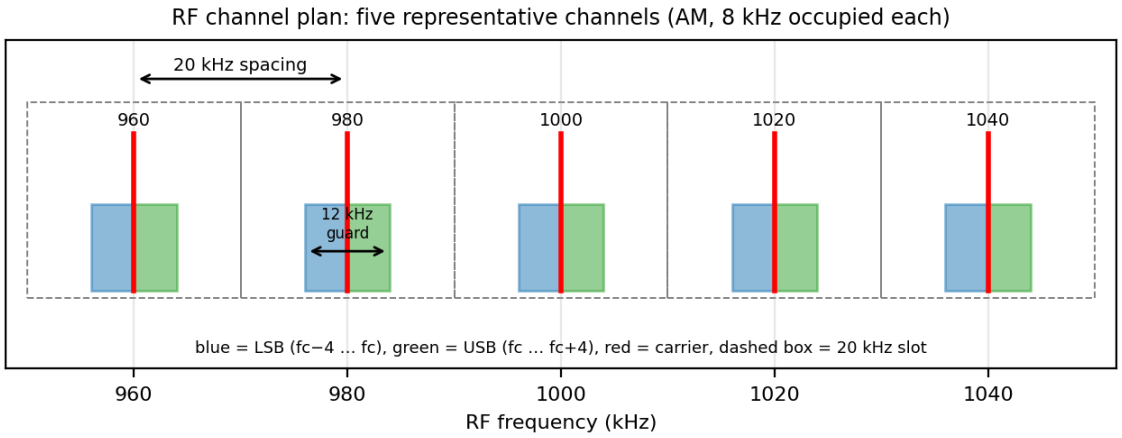


Fig. 4 RF channel plan: carriers, sidebands, 20 kHz slots and guard bands

3.2 Spacing, guard bands and overlap

- **Overlap check:** occupied bandwidth 8 kHz < spacing 20 kHz, so no overlap; the guard band between occupied bands is 20 − 8 = 12 kHz (6 kHz each side of the slot edge).
- **Why not closer?** Real receivers have finite IF-filter skirts (≈40 dB at 16 kHz in Sec. 4), oscillators drift (a 0.1 % LC oscillator at 1 MHz drifts 1 kHz; a crystal ±10 ppm drifts ±10 Hz), and transmitter non-linearity or over-modulation ($\mu > 1$) creates splatter beyond ±4 kHz. The guard band absorbs these.
- **Heterodyne whistle:** the beat between adjacent carriers is 20 kHz, above the 4 kHz audio filter, so it is inaudible; a 10 kHz spacing would leave an audible 10 kHz tone.
- **Transmit audio filter:** a 4 kHz low-pass filter with ≥ 40 dB attenuation at 12 kHz keeps the sidebands within ±10 kHz of the carrier.

3.3 Transmitter architecture

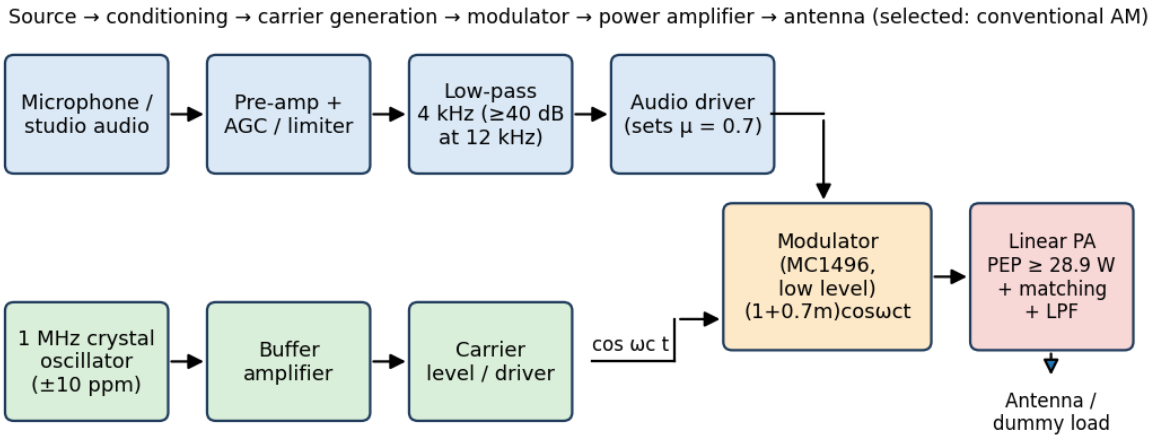


Fig. 5 Transmitter: source → conditioning → carrier generation → modulator → power amplifier → antenna

A crystal oscillator gives a stable 1 MHz carrier. The audio chain (pre-amp, AGC/limiter, 4 kHz low-pass filter) fixes the modulation depth at $\mu = 0.7$ and removes sidebands beyond 4 kHz. A **low-level modulator** (MC1496/AD633) is followed by a **linear class-AB/B power amplifier** (peak 28.9 W); this is simpler and safer for a laboratory than high-level collector modulation. The output filter removes harmonics before the antenna.

3.4 Generation and detection blocks, and VSB filtering

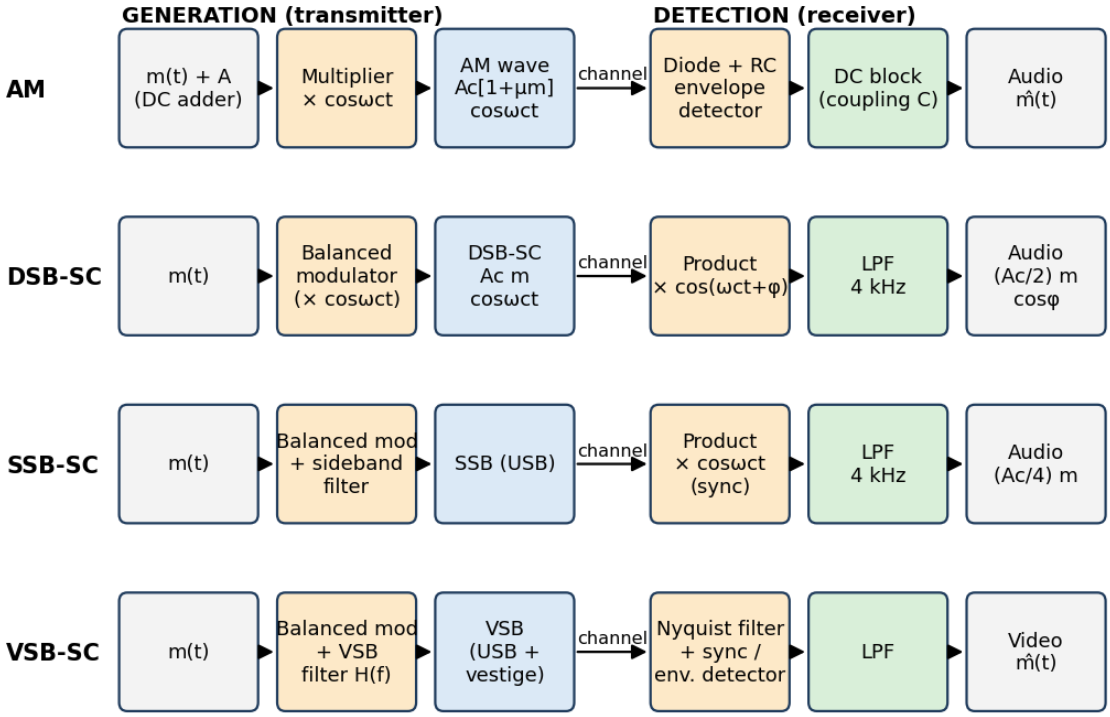


Fig. 6 Generation and detection blocks for AM, DSB-SC, SSB-SC and VSB-SC

VSB filtering concept: a DSB-SC signal is passed through a filter H(f) whose response rolls off smoothly through the carrier frequency, leaving one full sideband and a vestige of the other (Fig. 7). **Application:** analog television, where the 4.2 MHz video fits a 6 MHz channel together with the FM sound carrier at +4.5 MHz.

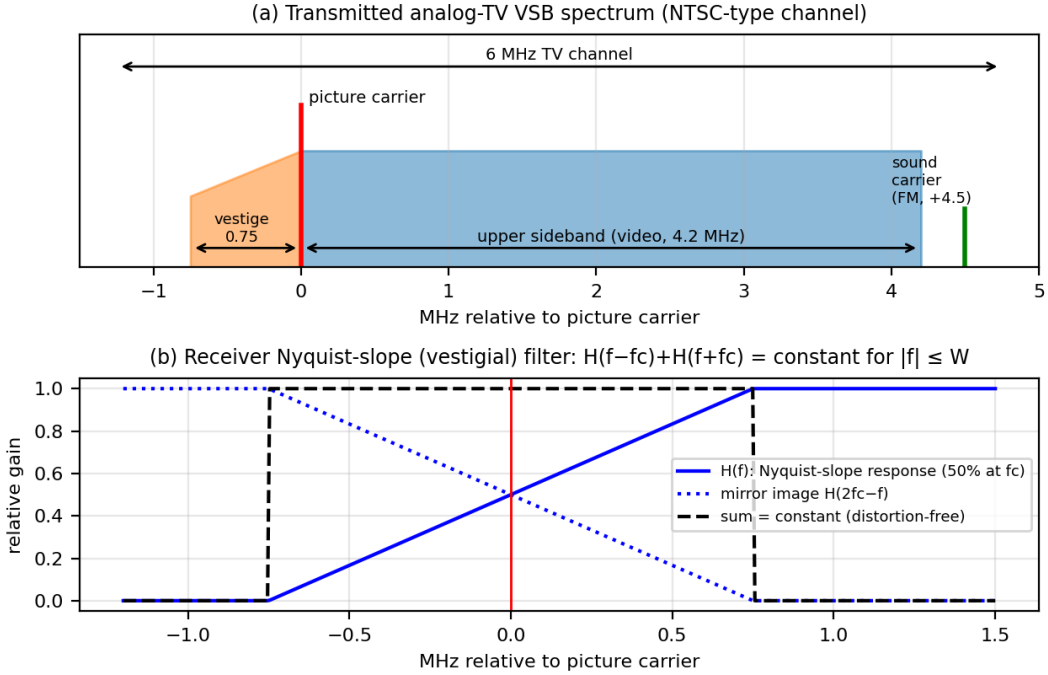


Fig. 7 VSB for analog TV: transmitted channel spectrum and the receiver Nyquist-slope filter (illustrative NTSC-type values)

3.5 Spectrum utilisation

Scheme	Occupied BW	Utilisation at 20 kHz spacing	Minimum practical spacing	Channels in 1060 kHz
AM / DSB-SC	8 kHz	40 %	≈ 10 kHz (8 + 2 guard)	53 at 20 kHz; ≈ 106 at 10 kHz
SSB-SC	4 kHz	20 %	≈ 5 kHz	≈ 212
VSB-SC	≈ 5 kHz	25 %	≈ 6 kHz	≈ 176

With the selected AM scheme, 52 channels × 8 kHz = 416 kHz of the 1060 kHz band carries signal (39 % utilisation); the rest is guard space that buys simple receivers.

Design Decision 2

Allocation: carriers on a 20 kHz grid anchored at 1 MHz (560 … 1580 kHz, 52 channels; 53 if the grid is shifted 10 kHz), each occupying ±4 kHz; design channel 1000 kHz. **Architecture:** crystal carrier, audio conditioning with 4 kHz low-pass filter, low-level AM modulator, linear PA (PEP 28.9 W), harmonic filter, antenna.

Trade-off: SSB would raise capacity to ≈ 212 channels but needs a steep sideband filter and coherent receivers; AM gives a third of that capacity but the simplest receivers. For a low-power educational broadcast link, simplicity wins.

Stage 4 – Select the AM Receiver Architecture

4.1 TRF and superheterodyne receivers

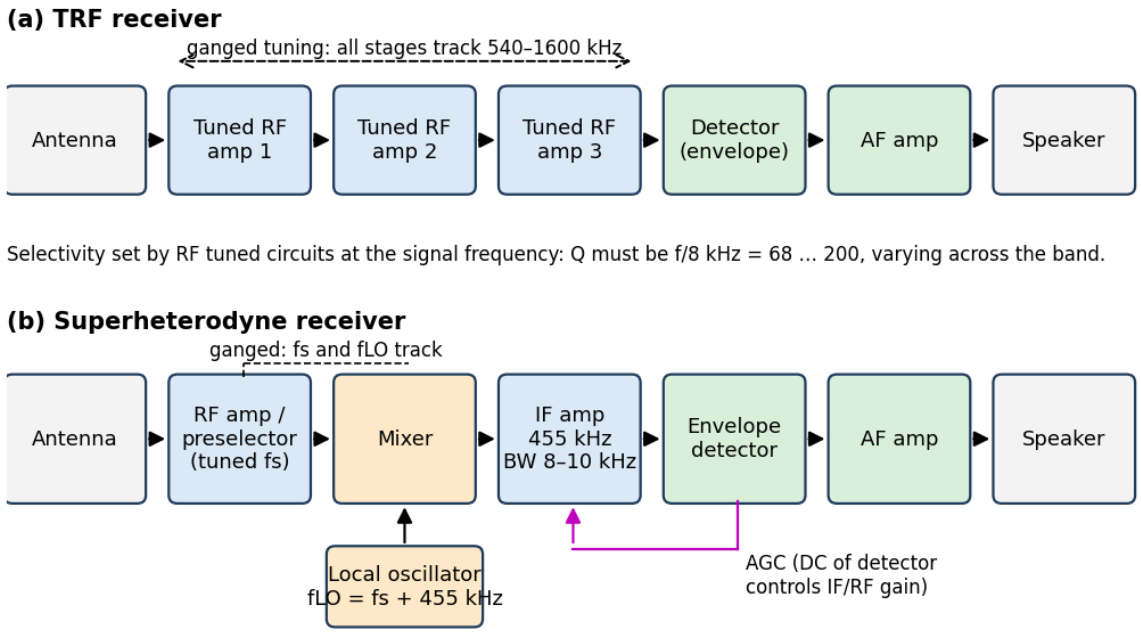


Fig. 8 (a) Tuned-radio-frequency receiver, (b) superheterodyne receiver

TRF: several tuned RF amplifiers, all tuned to the incoming frequency, then detector and audio amplifier. **Superheterodyne:** the preselector and RF amplifier give coarse selection and image rejection; the mixer with a local oscillator (f_LO = f_s + f_IF) translates **every** station to the fixed IF; the IF amplifier and filter give gain and sharp selectivity at one frequency; the detector recovers audio (AGC holds the level); the AF amplifier drives the speaker.

4.2 Comparison

Parameter	TRF	Superheterodyne
Selectivity	Needs Q = f/8 kHz = 68 (540 kHz) to 200 (1600 kHz). With Q = 100 the 3 dB bandwidth is 5.4 / 10.0 / 16.0 kHz at 540 / 1000 / 1600 kHz, so audio is cut at the low end and adjacent channels pass at the high end; one stage gives only 17.5 / 12.3 / 8.6 dB at ±20 kHz	Fixed IF filter (Q = 455/8 ≈ 57): same bandwidth for every station; 48.9 dB at ±20 kHz and 39.9 dB at ±16 kHz (simulated, 4th-order band-pass)

Parameter	TRF	Superheterodyne
Sensitivity	Limited; high gain at RF is prone to oscillation	High; most gain is at stable, low IF
Fidelity	Bandwidth varies with frequency	Constant IF bandwidth, 0.69 dB droop at 4 kHz
Image rejection	Not applicable (no image)	Needs RF preselector
Adjacent-channel rejection	Poor, varies across the band	Excellent, constant
Oscillator stability	No oscillator	LO drift detunes IF; needs stable LO
Tracking / complexity	Several ganged stages	Ganged RF + LO tracking; more parts but fewer high-Q stages

4.3 Detection

- **Envelope detection (AM):** output follows 1 + μ m(t). To avoid diagonal clipping at μ = 0.7 and f_m = 4 kHz: RC ≤ √(1−μ²)/(μω_m) = 0.714/(0.7 × 2π × 4000) = **40.6 μs**; and RC ≫ 1/f_IF = 2.2 μs. Choose R = 10 kΩ, C = 2.2 nF (RC = 22 μs). A coupling capacitor removes the DC and AGC takes the DC level.
- **Coherent detection (DSB-SC/SSB-SC):** multiply by a local carrier cos(ω_c t + φ) and low-pass filter: output (A_c/2) m(t) cosφ. A 30° error costs 1.25 dB; 90° gives zero; a frequency error Δf gives a beat (DSB) or a pitch shift of Δf (SSB). A Costas loop or PLL is needed.
- **VSB-TV receiver:** the IF filter has a Nyquist-slope shape that places the picture carrier at the 50 % point, with the sound carrier trapped separately; detection is by envelope or synchronous detector, the latter giving less quadrature distortion.

4.4 Selection of IF and image-frequency calculation

$$f_image = f_s + 2f_IF; \quad image\ rejection\ IR = \sqrt{(1 + Q^2\rho^2)},\ \rho = f_image/f_s - f_s/f_image$$

IF candidate	Image (f_s = 1 MHz)	ρ	IR, single stage Q = 50	Q needed for 40 dB	IF Q for 8 kHz BW
175 kHz	1350 kHz	0.61	29.7 dB	164.1	21.9
455 kHz	**1910 kHz**	1.39	**36.8 dB**	72.1	56.9
10.7 MHz	22400 kHz	22.36	61.0 dB	4.5	1338 (crystal filter)

Selected IF = 455 kHz: a low IF (175 kHz) gives a poor image ratio and puts the image inside the band; a high IF (10.7 MHz) gives excellent image rejection but needs a crystal filter (Q ≈ 1340) to get 8 kHz selectivity and costs far more; 455 kHz is a standard, cheap ceramic-filter frequency (Q = 57) that is outside the 540–1600 kHz band, so the IF itself cannot be picked up as a station.

Station f_s	f_LO = f_s + 455	Image f_s + 910	Image in broadcast band?	IR, one tuned stage (Q = 50)
540 kHz	995 kHz	1450 kHz	Yes (inside 540–1600)	41.3 dB
1000 kHz	**1455 kHz**	**1910 kHz**	No (above 1600 kHz)	**36.8 dB**
1600 kHz	2055 kHz	2510 kHz	No	33.4 dB (worst case)

Stations below 690 kHz have an image **inside** the band, so a tuned preselector is essential. LO tracking range is 995–2055 kHz (ratio 2.07, capacitance ratio 4.27), well within a standard gang capacitor. For the 1 MHz design channel the image at **1910 kHz** lies in the 160 m amateur band, so strong signals there are a real risk and the preselector must reject them by ≥ 36.8 dB (about 73 dB at 1 MHz and 67 dB at 1600 kHz with two tuned stages).

4.5 Receiver-parameter analysis

Parameter	Engineering discussion and number
Sensitivity	For 20 dB output SNR with AM at μ = 0.7 the channel SNR must be 20 dB + 7.06 dB = 27.1 dB (referred to N0·W). With NF = 10 dB, W = 4 kHz, 290 K: minimum input ≈ -100.9 dBm ≈ 2.0 μV rms across 50 Ω (thermal floor in an 8 kHz IF: -134.9 dBm).
Selectivity	IF filter: −39.9 dB at 16 kHz and −48.9 dB at 20 kHz off-centre; preselector adds 6.9 dB at 20 kHz, total 55.8 dB (simulated).
Fidelity	IF passband droop at 4 kHz is 0.7 dB; group delay about 85.2 μs (not audible); recovered-signal SDR 31.0 dB with no interference.

Parameter	Engineering discussion and number
Image rejection	36.8 dB at 1 MHz with Q = 50 (simulation equals formula); worst case 33.4 dB at 1600 kHz.
IF stability	LO error shifts the IF one-for-one. A 0.1 % LC oscillator (1.45 kHz) still keeps the signal inside the ±5 kHz IF window but reduces adjacent-channel margin; a PLL/crystal-referenced LO is recommended.
Detection	Diode envelope detector (RC = 22 µs) matches AM; coherent detection would add a carrier-recovery loop for no benefit with a transmitted carrier.

Design Decision 3

I recommend the **superheterodyne receiver, IF = 455 kHz, f_LO = f_s + 455 kHz (1455 kHz for 1000 kHz), envelope detector with AGC**, because: (i) one fixed 8–10 kHz IF filter (Q ≈ 57) gives the same bandwidth and ≥ 40 dB rejection at ±16 kHz for every station, whereas a TRF would need Q = 68–200 tracking over the band; (ii) image 1910 kHz is rejected by 36.8 dB with one Q = 50 stage; (iii) the cost is only LO tracking and a stable LO, both standard.

Stage 5 – Develop, Verify and Compare the Complete System

5.1 End-to-end block diagram and spectra at key points

Complete end-to-end AM link: source → transmitter → channel → superheterodyne receiver → destination

Test points A–F correspond to the spectra sketched below and the simulated spectra in Section 5.

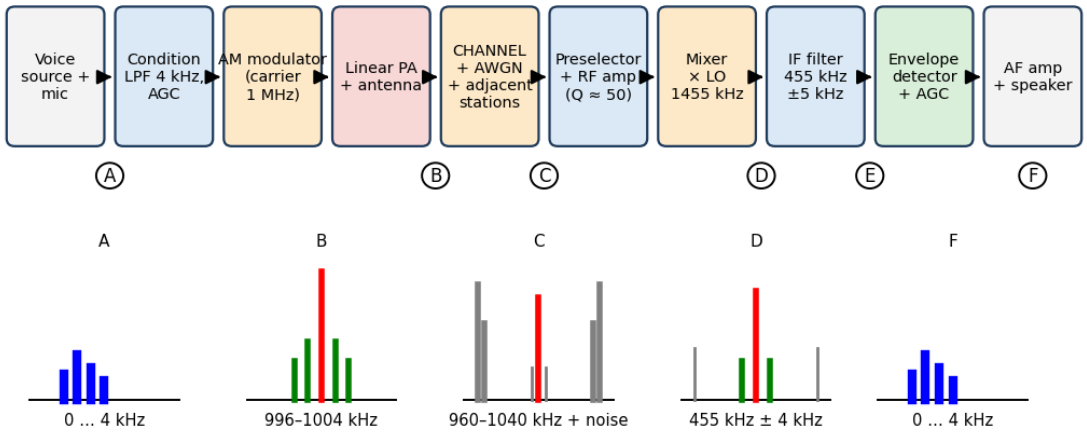


Fig. 9 Complete end-to-end AM link with test points A–F

5.2 Verification of power and bandwidth (simulation, 1 Ω reference)

Quantity	Calculated	Simulated (Python, 1 kHz tone)
Modulation index µ	0.700	0.70
Carrier power P_c	10.00 W	10.00 W
LSB power / USB power	1.225 W each	1.225 W / 1.225 W
Sideband total	2.45 W	2.45 W
Total power P_T	12.45 W	12.45 W
Peak envelope power	28.9 W	28.9 W
Occupied bandwidth	8 kHz	8 kHz (±4 kHz around 1 MHz, Fig. 3a)

The simulation agrees with the analytical results to numerical precision, confirming $P_T = P_c(1 + \mu^2/2)$ and that only 19.7 % of the power sits in the sidebands.

5.3 Receiver simulation: selectivity, image rejection and fidelity

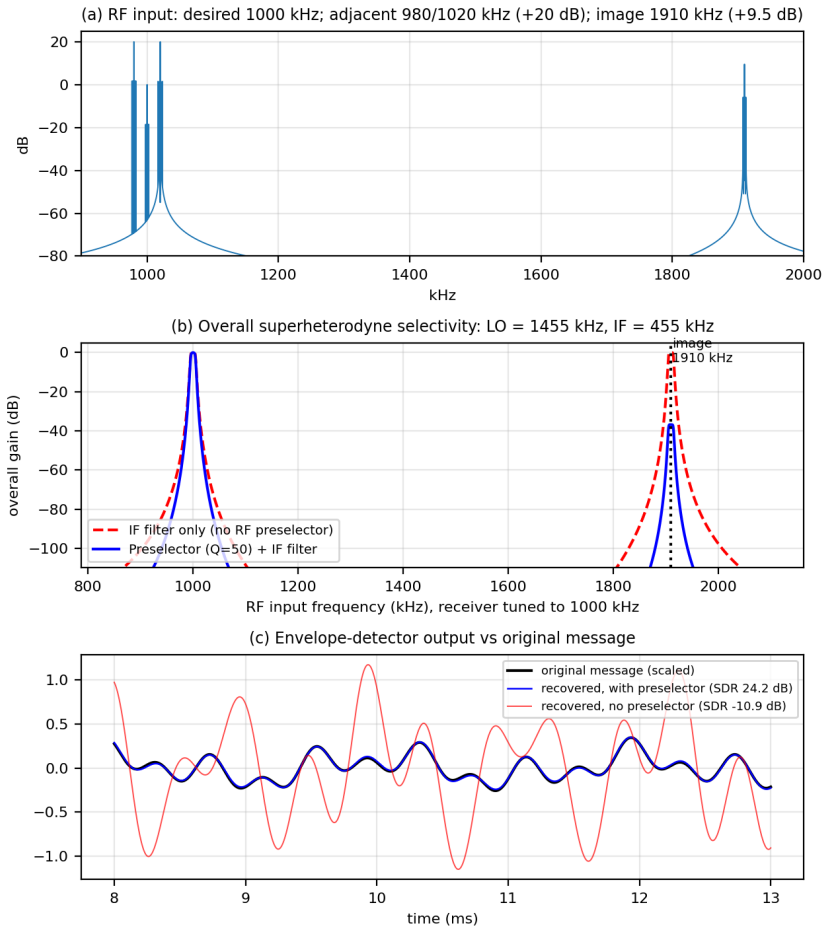


Fig. 10 Receiver simulation: RF input, overall selectivity and recovered audio

Interpretation: the receiver is tuned to 1000 kHz (f_LO = 1455 kHz) with adjacent stations at 980 and 1020 kHz 20 dB stronger and an image interferer at 1910 kHz 9.5 dB stronger than the wanted signal. With the preselector the recovered audio matches the original with an SDR of **24.2 dB** (31.0 dB with no interferers); without the preselector the image passes straight through the IF filter (0 dB) and the SDR collapses to **-10.9 dB**. This shows that the IF filter alone gives selectivity but not image protection; both are required.

5.4 Noise effects and comparison with rejected alternatives

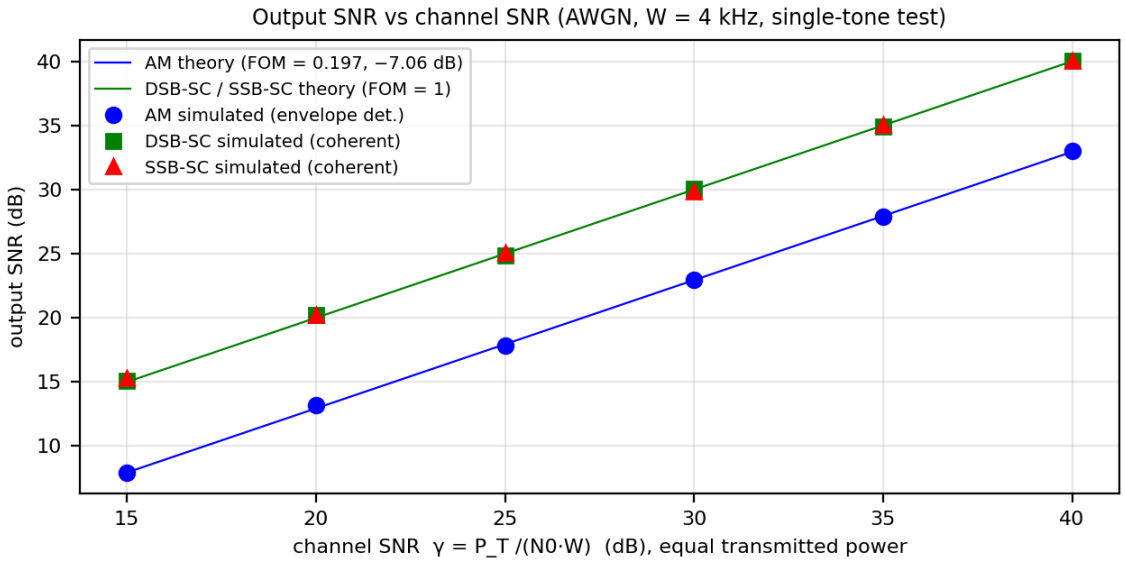


Fig. 11 Output SNR versus channel SNR at equal transmitted power (AWGN)

Channel SNR γ (dB)	15	20	25	30	35	40
AM envelope, output SNR (dB)	8.0	13.2	17.8	23.0	27.9	33.0
DSB-SC coherent (dB)	15.1	20.3	24.9	30.1	34.9	40.1
SSB-SC coherent (dB)	15.3	20.2	25.1	29.9	35.1	40.1

Interpretation: at equal transmitted power AM is 7.06 dB worse than DSB-SC or SSB-SC, as predicted by the figure of merit $\mu^2/(2+\mu^2) = 0.197$; the simulated values are within about 0.3 dB of theory. The envelope detector also has a **threshold** (below roughly 10 dB carrier-to-noise ratio in the 8 kHz IF the output degrades abruptly); coherent detectors do not. A narrower IF reduces noise power, which is why selectivity improves SNR as well as interference rejection. For the voice link, 20 dB output SNR needs $\gamma \approx 27$ dB with AM, i.e. 7 dB more power than DSB-SC – the price paid for the simple receiver.

Rejected alternative – SSB-SC: halves the bandwidth (4 vs 8 kHz) and needs only 1.225 W instead of 12.45 W for the same output SNR (10.1 dB less power), but requires carrier-synchronous detection (pitch error equal to the LO error), a very steep sideband filter and a more expensive receiver. **DSB-SC** saves 80 % of the power but has the same 8 kHz bandwidth as AM, so it pays for coherent detection without any spectrum gain.

5.5 Suitability for a low-cost laboratory implementation

AM with a superheterodyne receiver is the lowest-cost option: MC1496/AD633 modulator, crystal oscillator, 455 kHz ceramic filter and a diode detector are all inexpensive; all waveforms can be checked with an oscilloscope and sound-card spectrum analyser. SSB/VSB would need precision filters and a synchronous detector.

Stage 6 – Critical Evaluation and Deployment Decision

6.1 Comparison of the four schemes

Scheme	BW (kHz)	P_T for same sidebands	Efficiency	Output SNR at equal P_T	Receiver
AM	8	12.45 W	19.7 %	−7.06 dB	Envelope detector (simplest)
DSB-SC	8	2.45 W	100 %	0 dB	Coherent, carrier recovery
SSB-SC	4	1.225 W	100 %	0 dB	Coherent, steep filter
VSB-SC	≈ 5	≈ 1.2–2.5 W	high	≈ 0 dB	Nyquist filter + sync/envelope

6.2 Suitability: voice broadcasting versus analog TV

- **Voice broadcasting:** suitable. 8 kHz fits the 20 kHz slot; listeners' receivers are cheap; $\mu = 0.7$ gives good depth without distortion; the 4 kHz audio band is adequate for speech (not for high-fidelity music).
- **Analog TV:** not suitable with this chain. Video needs 4.2 MHz, so a 1 MHz carrier and 20 kHz spacing are impossible; AM/DSB would need 8.4 MHz, so VSB (6 MHz channel, 0.75 MHz vestige, Nyquist-slope receiver filter, carrier at 50 % point) is required. The vestige's gradual roll-off, however, adds quadrature distortion and needs precise filters, which is the VSB-TV limitation.

6.3 Final specification verification

Requirement	Target	Result	Status
Message bandwidth	4 kHz	4 kHz audio low-pass filter	Met
Occupied bandwidth	≤ 20 kHz slot	8 kHz (40 %), 12 kHz guard	Met
Carrier / band	1 MHz in 540–1600 kHz	1000 kHz, 52–53 channels	Met
Modulation index	0.7	0.700 (simulated)	Met
Carrier power	10 W	10.00 W; P_T 12.45 W; sidebands 2.45 W	Met
PA rating	Linear at peak	PEP 28.9 W	Specified
Image rejection	≥ 30 dB	36.8 dB at 1 MHz (33.4 dB worst case)	Met
Adjacent-channel rejection	≥ 40 dB at ±16 kHz	39.9 dB at 16 kHz (IF filter alone, ≈ 45 dB with preselector); 55.8 dB overall at 20 kHz	Met

Requirement	Target	Result	Status
Sensitivity	20 dB output SNR	≈ 2.0 µV (NF 10 dB)	Met (calculated)
Fidelity	< 1 dB droop at 4 kHz	0.7 dB; SDR 31.0 dB	Met
Detector	No diagonal clipping	RC = 22 µs ≤ 40.6 µs	Met
Power efficiency	–	19.7 %; AM output SNR 7.06 dB below DSB/SSB	Limitation

FINAL ENGINEERING DECISION – WOULD YOU DEPLOY THIS COMMUNICATION SYSTEM? YES

Decision: YES – deploy as a **voice-broadcast** link (and as the educational prototype).

Defence: bandwidth (8 kHz ≤ 20 kHz slot) and power (P_T = 12.45 W, µ = 0.7) comply; the 20 kHz plan has no overlap (12 kHz guard) and 52–53 channels; the receiver is a simple diode detector; with IF = 455 kHz the image (1910 kHz) is rejected by 36.8 dB and adjacent channels by ≥ 40 dB; cost is minimal; the measured/simulated figures all match the calculations.

Limits that I accept: (1) only 19.7 % of the power is information; (2) output SNR is 7.06 dB below DSB/SSB at equal power; (3) 40 % spectrum utilisation; (4) stations below 690 kHz have an in-band image, so the preselector is critical; (5) the chain is **not** valid for analog TV, which needs VSB and a 6 MHz channel.

If requirements change (redesign): for a power-limited or spectrum-limited service, switch to **SSB-SC with a pilot carrier** (power 12.45 W → 1.225 W, spacing 20 → 5 kHz, ≈ 212 channels) and a coherent receiver with a 455 kHz crystal filter; for TV use VSB with a 6 MHz plan, 45.75 MHz-class IF and a Nyquist-slope filter. For better selectivity add a second tuned RF stage (image rejection ≈ 67 dB worst case) and a crystal-referenced LO.